SECURECORP

Enterprise Network & Cybersecurity Lab

Cisco Packet Tracer | Multi-site routing | VLAN segmentation | ACLs | SSH | NAT/PAT | DMZ

Technical implementation and evidence portfolio | October 2026

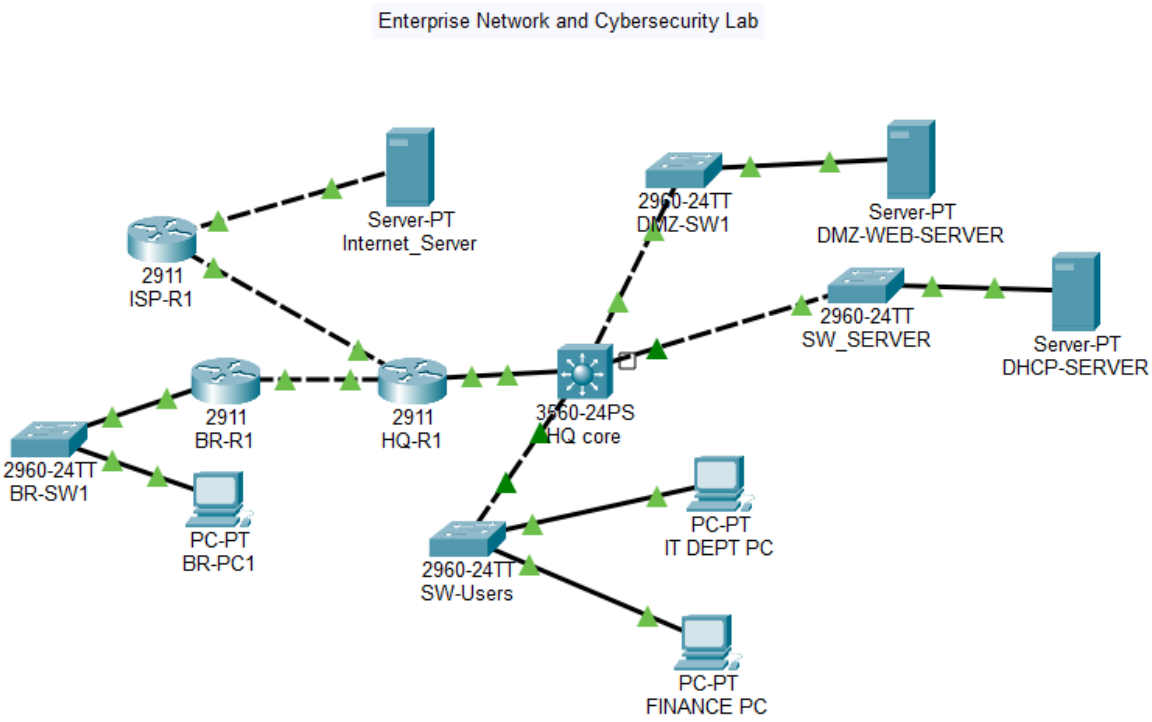


Figure 1. Complete logical network topology, showing HQ, Branch, DMZ, and simulated ISP.

1. Executive Summary

SecureCorp is a Cisco Packet Tracer enterprise network simulation with a headquarters campus, remote branch office, simulated Internet service provider, and demilitarized zone (DMZ). The lab integrates Layer 2 segmentation, Layer 3 routing, dynamic routing, centralized DHCP, remote administration controls, switch port security, NAT/PAT, and a published HTTP service.

The build was validated using CLI configuration evidence, routing tables, ping tests, SSH access tests, and an external browser test. A remaining DMZ isolation discrepancy is explicitly recorded in the limitations section rather than represented as resolved.

2. Architecture and Addressing

Zone / VLAN	Subnet	Gateway / purpose
IT / VLAN 10	10.10.10.0/24	10.10.10.1
Finance / VLAN 20	10.10.20.0/24	10.10.20.1
Internal servers / VLAN 30	10.10.30.0/24	10.10.30.1
Branch users / VLAN 40	10.20.10.0/24	10.20.10.1 (BR-R1)
DMZ / VLAN 50	10.10.50.0/24	10.10.50.1
HQ core-router transit	10.255.0.0/30	HQ-CORE ↔ HQ-R1
HQ-Branch WAN	10.255.1.0/30	HQ-R1 ↔ BR-R1
Simulated ISP transit	203.0.113.0/30	HQ-R1 ↔ ISP-R1
External test network	198.51.100.0/24	INTERNET-SERVER 198.51.100.10

HQ-CORE (Cisco 3560) performs routing for HQ VLANs; BR-R1 routes the Branch LAN. HQ-R1 provides the inter-site WAN connection and simulated Internet edge. Documentation-only TEST-NET ranges represent the outside network, not access to the public Internet.

```
[OK]
HQ-CORE#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Gig0/2
10   IT                               active
20   finance                          active
30   SERVERS                          active
50   DMZ                              active    Fa0/22
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

Figure 2. HQ-CORE VLAN inventory: IT, Finance, SERVERS, and DMZ.

```
HQ-CORE#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES unset  down                  down
FastEthernet0/2        unassigned      YES unset  down                  down
FastEthernet0/3        unassigned      YES unset  down                  down
FastEthernet0/4        unassigned      YES unset  down                  down
FastEthernet0/5        unassigned      YES unset  down                  down
FastEthernet0/6        unassigned      YES unset  down                  down
FastEthernet0/7        unassigned      YES unset  down                  down
FastEthernet0/8        unassigned      YES unset  down                  down
FastEthernet0/9        unassigned      YES unset  down                  down
FastEthernet0/10       unassigned      YES unset  down                  down
FastEthernet0/11       unassigned      YES unset  down                  down
FastEthernet0/12       unassigned      YES unset  down                  down
FastEthernet0/13       unassigned      YES unset  down                  down
FastEthernet0/14       unassigned      YES unset  down                  down
FastEthernet0/15       unassigned      YES unset  down                  down
FastEthernet0/16       unassigned      YES unset  down                  down
FastEthernet0/17       unassigned      YES unset  down                  down
FastEthernet0/18       unassigned      YES unset  down                  down
FastEthernet0/19       unassigned      YES unset  down                  down
FastEthernet0/20       unassigned      YES unset  down                  down
FastEthernet0/21       unassigned      YES unset  down                  down
FastEthernet0/22       unassigned      YES unset  up                    up
FastEthernet0/23       unassigned      YES unset  up                    up
FastEthernet0/24       unassigned      YES unset  up                    up
GigabitEthernet0/1     10.255.0.1      YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  down                  down
Vlan1                  unassigned      YES unset  administratively down down
Vlan10                 10.10.10.1      YES manual up                    up
Vlan20                 10.10.20.1      YES manual up                    up
Vlan30                 10.10.30.1      YES manual up                    up
Vlan50                 10.10.50.1      YES manual up                    up
```

Figure 3. HQ-CORE switched virtual interfaces and routed uplink.

3. Centralized DHCP and Relay

DHCP-SERVER (10.10.30.10) supplies separate pools to IT and Finance. HQ-CORE relays client DHCP traffic from VLAN 10 and VLAN 20 with **ip helper-address 10.10.30.10**.

Pool name	Starting address	Gateway	Configured maximum
IT_DEPARTMENT	10.10.10.100	10.10.10.1	156
financepool	10.10.20.100	10.10.20.1	156

Configuration note: Starting at .100 with a maximum of 156 reaches .255, the subnet broadcast address. A maximum of 155 would end at the last usable host (.254); pool sizing should be reviewed.

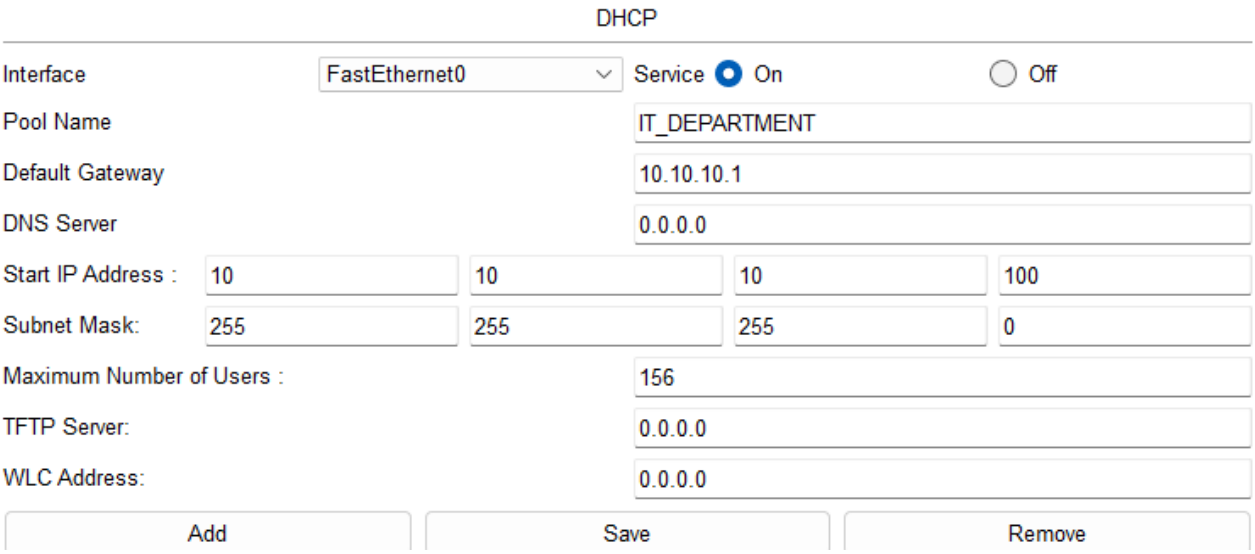


Figure 4. IT_DEPARTMENT DHCP pool.

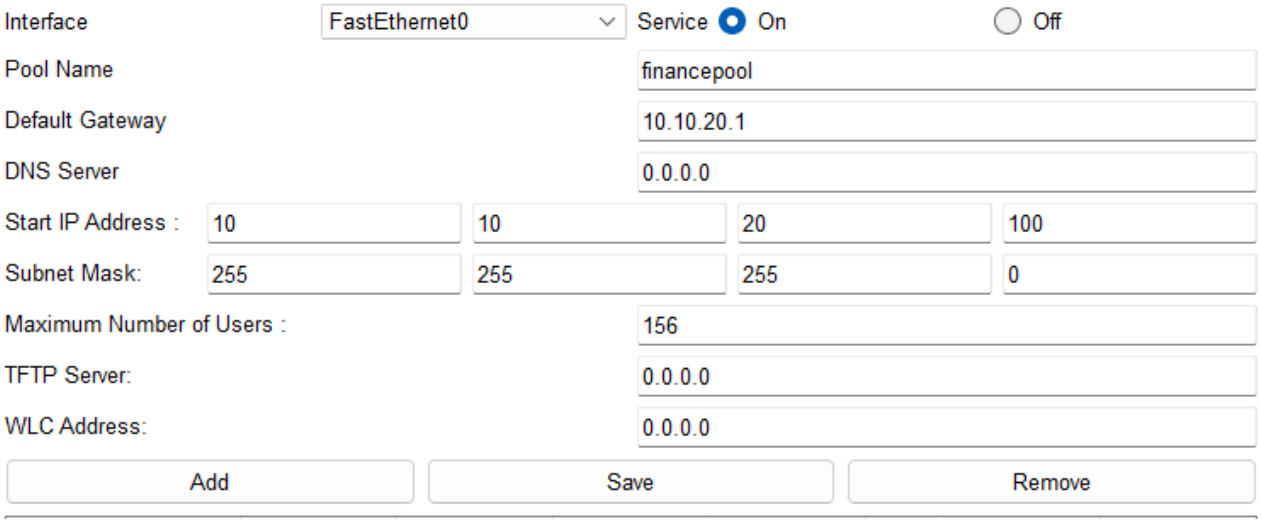


Figure 5. financepool DHCP pool.

```
interface Vlan10
 mac-address 00e0.f794.0401
 ip address 10.10.10.1 255.255.255.0
 ip helper-address 10.10.30.10
!
```

Figure 6A. VLAN 10 DHCP relay configuration.

```
interface Vlan20
 mac-address 00e0.f794.0402
 ip address 10.10.20.1 255.255.255.0
 ip helper-address 10.10.30.10
 ip access-group Finance_Security in
!
```

Figure 6B. VLAN 20 DHCP relay and inbound Finance ACL.

4. OSPF and Branch Connectivity

OSPF process 1, area 0, runs on HQ-R1 (router ID 1.1.1.1), BR-R1 (2.2.2.2), and HQ-CORE (3.3.3.3). HQ-R1 advertises its installed default route using **default-information originate**. The Branch routing table shows HQ subnets learned via 10.255.1.1 and an O*E2 default route.

```
Router#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           1   FULL/DR         00:00:31    10.255.1.2      GigabitEthernet0/0
3.3.3.3           1   FULL/DR         00:00:33    10.255.0.1      GigabitEthernet0/1
```

Figure 7. HQ-R1 OSPF neighbors in FULL/DR state.

```
BR-R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.255.1.1 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 9 subnets, 3 masks
O       10.10.10.0/24 [110/3] via 10.255.1.1, 00:25:09, GigabitEthernet0/1
O       10.10.20.0/24 [110/3] via 10.255.1.1, 00:25:09, GigabitEthernet0/1
O       10.10.30.0/24 [110/3] via 10.255.1.1, 00:25:09, GigabitEthernet0/1
O       10.10.50.0/24 [110/3] via 10.255.1.1, 00:25:09, GigabitEthernet0/1
C       10.20.10.0/24 is directly connected, GigabitEthernet0/0
L       10.20.10.1/32 is directly connected, GigabitEthernet0/0
O       10.255.0.0/30 [110/2] via 10.255.1.1, 00:25:09, GigabitEthernet0/1
C       10.255.1.0/30 is directly connected, GigabitEthernet0/1
L       10.255.1.2/32 is directly connected, GigabitEthernet0/1
O*E2 0.0.0.0/0 [110/1] via 10.255.1.1, 00:25:09, GigabitEthernet0/1
```

Figure 8. BR-R1 dynamically learned HQ routes and OSPF default route.

```
router ospf 1
 router-id 1.1.1.1
 log-adjacency-changes
 network 10.255.0.0 0.0.0.3 area 0
 network 10.255.1.0 0.0.0.3 area 0
 default-information originate
```

Figure 9. HQ-R1 OSPF network statements and default route origination.

5. Security Controls

5.1 Finance segmentation

The Finance_Security extended ACL is applied inbound on VLAN 20. It permits DHCP client requests, denies Finance-to-IT traffic, and permits remaining Finance traffic. The deny entry recorded 16 matches; testing confirmed Finance could reach the internal DHCP server while failing to reach IT-PC.

```
HQ-CORE# show access-lists
Extended IP access list Finance_Security
    10 permit udp any eq bootpc any eq bootps (3 match(es))
    20 deny ip 10.10.20.0 0.0.0.255 10.10.10.0 0.0.0.255 (16 match(es))
    30 permit ip 10.10.20.0 0.0.0.255 any (4 match(es))
```

Figure 10. Finance_Security ACL and packet counters.

5.2 SSH administration

HQ-CORE uses SSH v2 and local administrator authentication. The SSH_MANAGEMENT standard ACL allows management connections from the IT subnet (10.10.10.0/24) and denies other sources. Finance SSH timed out; IT SSH succeeded.

```
    30 permit ip 10.10.20.0 0.0.0.255 any (4 match(es))
Standard IP access list SSH_MANAGEMENT
    10 permit 10.10.10.0 0.0.0.255 (2 match(es))
    20 deny any
```

Figure 11. SSH_MANAGEMENT source-IP restrictions.

5.3 Access-port security

SW-USERS FastEthernet0/1 uses sticky MAC learning, a maximum of one secure MAC address, and restrict violation mode. The captured interface was Secure-up with zero current violations.

```
SW_USERS#show port-security interface fastEthernet 0/1
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Restrict
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 1
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 1
Last Source Address:Vlan   : 0030.A338.8DAB:10
Security Violation Count   : 0
```

Figure 12. IT access-port security operational state.

5.4 DMZ ACL

The DMZ_SECURITY extended ACL is configured inbound on VLAN 50, with denies from 10.10.50.0/24 to protected internal subnets and a TCP established-flag exception for HTTP responses to IT. The established keyword is a TCP flag check, not stateful connection tracking. A later ICMP test unexpectedly succeeded; see Known Limitations.

```
HQ-CORE# show access-lists DMZ_SECURITY
Extended IP access list DMZ_SECURITY
    permit tcp host 10.10.50.10 eq www 10.10.10.0 0.0.0.255 established
    deny ip 10.10.50.0 0.0.0.255 10.10.10.0 0.0.0.255 (4 match(es))
    deny ip 10.10.50.0 0.0.0.255 10.10.20.0 0.0.0.255
    deny ip 10.10.50.0 0.0.0.255 10.10.30.0 0.0.0.255
    deny ip 10.10.50.0 0.0.0.255 10.20.10.0 0.0.0.255
    permit ip 10.10.50.0 0.0.0.255 any (3 match(es))
```

Figure 13. DMZ_SECURITY ACL with deny and permit entries.

6. NAT/PAT and Public Web Service

HQ-R1 marks GigabitEthernet0/0 and GigabitEthernet0/1 as NAT inside, and GigabitEthernet0/2 as NAT outside. Dynamic PAT translates permitted internal source addresses to 203.0.113.2. Static PAT maps TCP 203.0.113.2:80 to

DMZ-WEB-SERVER 10.10.50.10:80.

```
Router#show ip nat statistics
Total translations: 2 (1 static, 1 dynamic, 2 extended)
Outside Interfaces: GigabitEthernet0/2
Inside Interfaces: GigabitEthernet0/0 , GigabitEthernet0/1
Hits: 12  Misses: 5
Expired translations: 4
Dynamic mappings:
```

Figure 14. HQ-R1 NAT statistics and interface roles.

```
Router#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
tcp 203.0.113.2:80     10.10.50.10:80     ---                ---
tcp 203.0.113.2:80     10.10.50.10:80     198.51.100.10:1025 198.51.100.10:1025
```

Figure 15. Static TCP port 80 mapping and external HTTP connection.



Figure 16. External browser successfully loads the DMZ website at http://203.0.113.2.

7. Verification Evidence

Test	Observed result
Finance → IT ping	Failed; gateway reported destination host unreachable
Finance → internal server	4/4 replies, 0% loss
Finance → HQ-CORE SSH	Connection timed out
IT → HQ-CORE SSH	Successful login to HQ-CORE#
Branch → HQ DHCP server	4/4 replies, 0% loss
Branch → external test server	4/4 replies, 0% loss
External client → DMZ HTTP	Webpage loaded through public static PAT
DMZ → IT ping (latest test)	Succeeded unexpectedly; isolation not verified

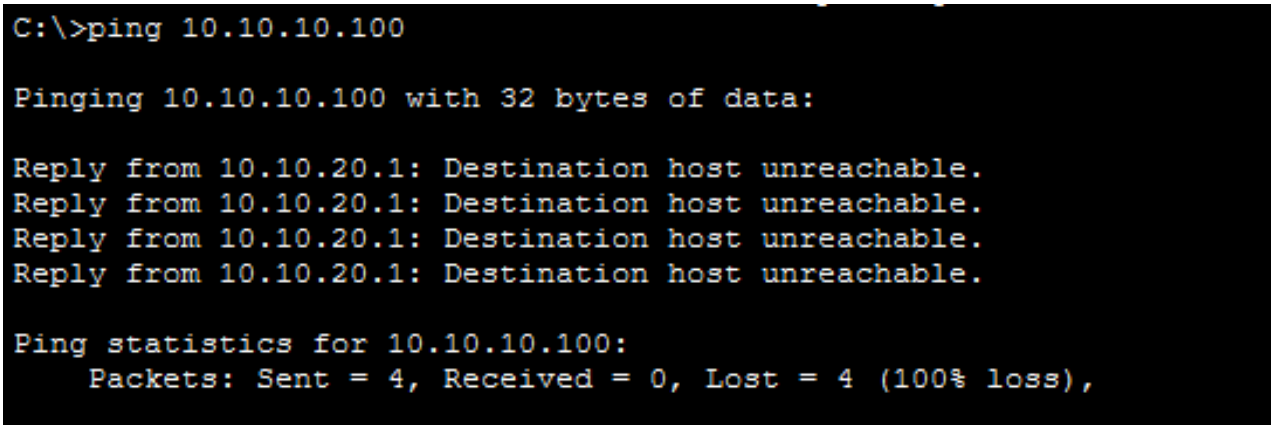


Figure 17. Finance-to-IT ping denied (4/4 lost).

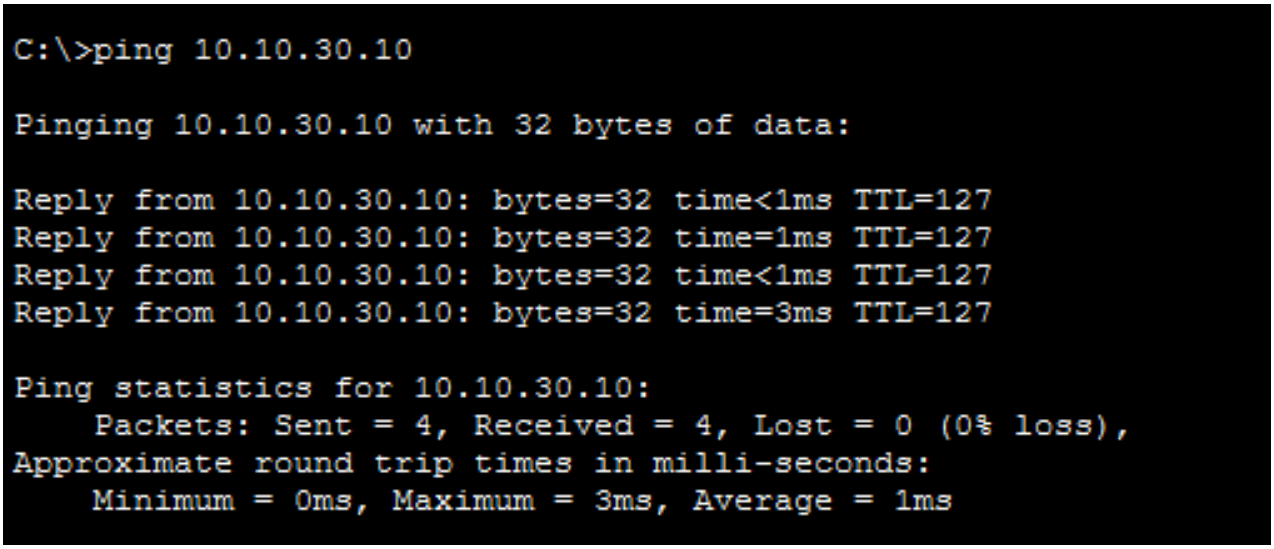


Figure 18. Finance-to-server ping allowed (4/4 replies).

```
C:\>ssh -l netadmin 10.10.10.1

% Connection timed out; remote host not responding
C:\>
```

Figure 19. Finance SSH management attempt times out.

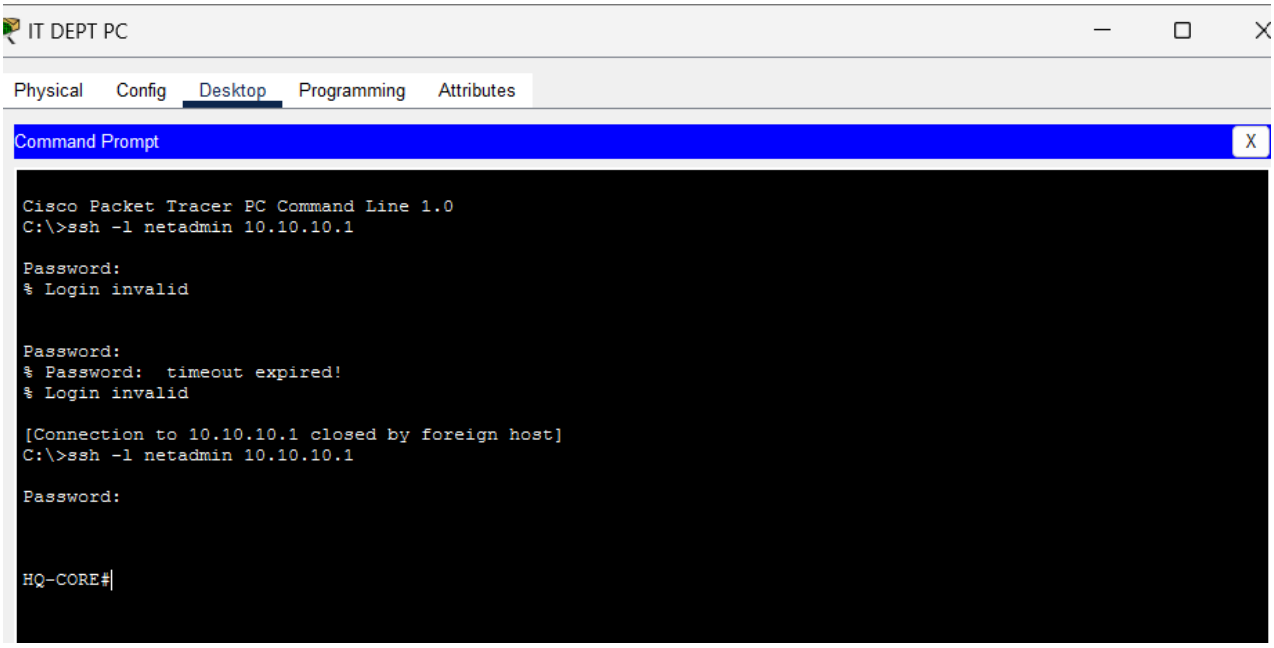


Figure 20. IT-PC authenticates successfully to HQ-CORE via SSH.

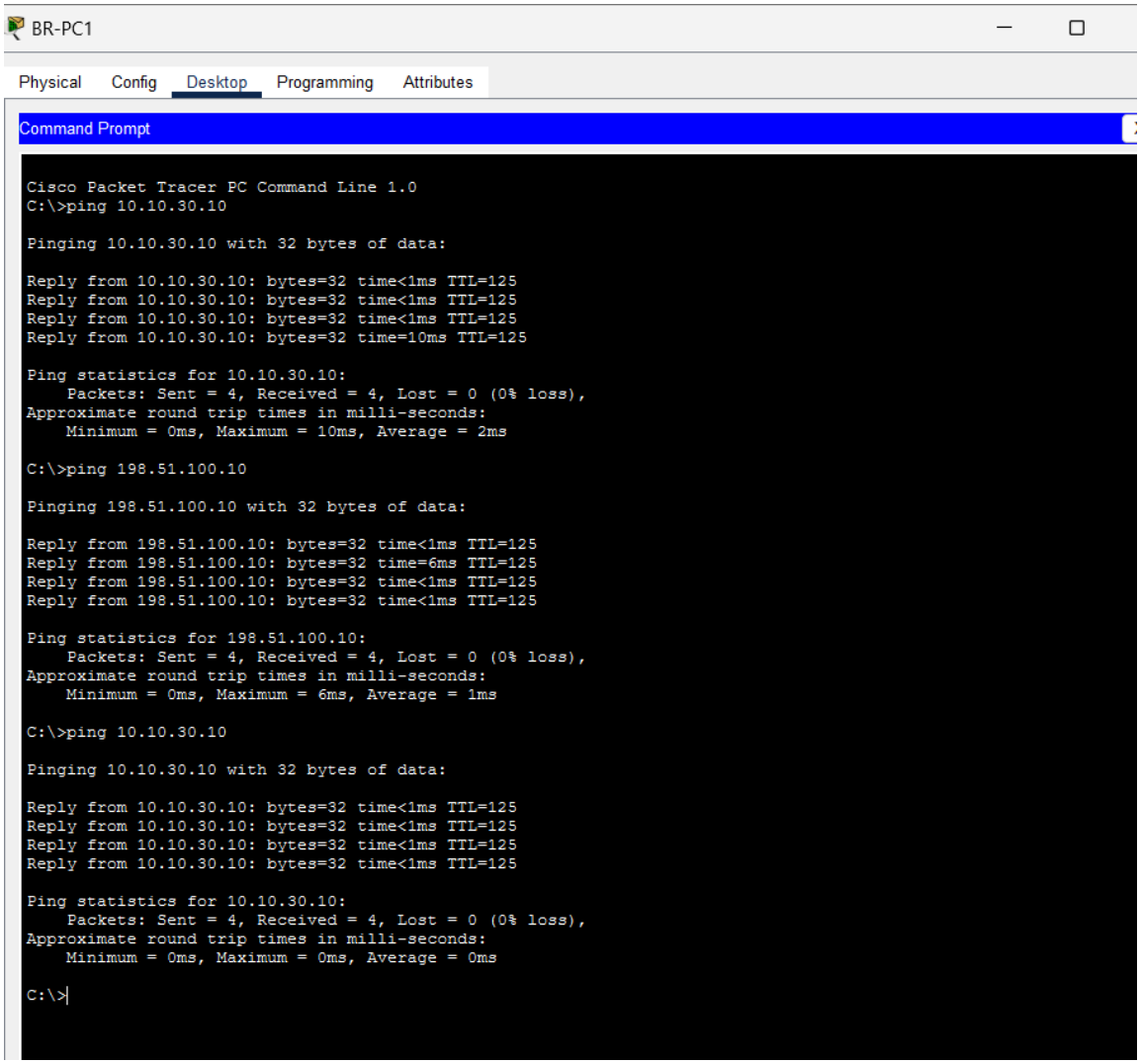


Figure 21. Branch pings to HQ and simulated Internet succeed.

8. Troubleshooting and Lessons Learned

OSPF: Duplicate router IDs prevented stable neighbor formation; unique IDs restored FULL adjacencies. **Branch routing:** A missing workstation default gateway was corrected. **NAT:** The missing outside interface role and a mistaken Branch source subnet in the NAT ACL were fixed. **ISP:** A mistyped interface IP was corrected. **Static PAT:** An unsupported interface-based syntax was replaced with an explicit outside address.

These incidents reinforced layered troubleshooting: verify addressing and gateways, inspect interface state and routes, check policy placement and counters, then validate application traffic end to end.

9. Known Limitation: DMZ Isolation

A later test from DMZ-WEB-SERVER (10.10.50.10) to IT-PC (10.10.10.100) received 4/4 replies despite the inbound VLAN 50 DMZ_SECURITY ACL containing a matching deny. The deny counter did not increase during that test. The server IP and gateway and the ACL interface assignment were verified. Root cause remains unresolved, so full DMZ isolation must not be claimed.

```
C:\>ping 10.10.10.100

Pinging 10.10.10.100 with 32 bytes of data:

Reply from 10.10.10.100: bytes=32 time<1ms TTL=125
Reply from 10.10.10.100: bytes=32 time<1ms TTL=125
Reply from 10.10.10.100: bytes=32 time<1ms TTL=125
Reply from 10.10.10.100: bytes=32 time=1ms TTL=125

Ping statistics for 10.10.10.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Figure 22. Unexpected successful DMZ-to-IT ping; unresolved security finding.

```
interface Vlan50
 description DMZ_GATEWAY
 mac-address 00e0.f794.0404
 ip address 10.10.50.1 255.255.255.0
 ip access-group DMZ_SECURITY in
!
```

Figure 22A. VLAN 50 running configuration shows inbound DMZ_SECURITY ACL.

```
HQ-CORE# show access-lists DMZ_SECURITY
Extended IP access list DMZ_SECURITY
    permit tcp host 10.10.50.10 eq www 10.10.10.0 0.0.0.255 established
    deny ip 10.10.50.0 0.0.0.255 10.10.10.0 0.0.0.255 (4 match(es))
    deny ip 10.10.50.0 0.0.0.255 10.10.20.0 0.0.0.255
    deny ip 10.10.50.0 0.0.0.255 10.10.30.0 0.0.0.255
    deny ip 10.10.50.0 0.0.0.255 10.20.10.0 0.0.0.255
    permit ip 10.10.50.0 0.0.0.255 any (3 match(es))
```

Figure 22B. DMZ ACL counters remain unchanged after the unexpected ping.

Future work: Trace packets in Packet Tracer Simulation Mode, verify the precise ingress path and forwarding behavior, review ACL evaluation and platform limitations, and repeat on a supported IOS environment. Consider a stateful firewall and an explicit least-privilege DMZ policy.

10. Project Outcome

SecureCorp demonstrates a functioning simulated multi-site enterprise network with routed VLANs, centralized DHCP, OSPF, source-restricted SSH administration, access-port security, centralized NAT/PAT, and an externally reachable DMZ HTTP service. Most connectivity and access-control tests passed. The unresolved DMZ-to-IT ping discrepancy is transparently documented as a security validation gap.

Suggested portfolio assets: SecureCorp_Final_Network.pkt, this PDF, and the evidence screenshots.